

WASP-43b

R-0508

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_230128 · created 2026-07-20T18:58:00+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459972.8744 +/- 0.0022 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.178 +/- 0.033
Transit depth (Rp/Rs)^2	3.18 +/- 1.18 [%]
Orbital Inclination (inc)	79.25 +/- 2.34
Airmass coefficient 1 (a1)	0.823 +/- 0.01
Airmass coefficient 2 (a2)	0.142 +/- 0.05
Scatter in the residuals of the lightcurve fit is	1.26 %
Best Comparison Star	#2 - [380, 133]
Optimal Aperture	4.39
Optimal Annulus	8.05
Transit Duration (day)	0.037 +/- 0.016

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.178 ± 0.033 vs 0.1594 (deviation 0.56σ)
Duration fit vs published	0.037 d vs 0.0512 d (deviation 0.89σ)
Mid-time O-C	-4.5 min
Depth SNR	5.4
Residual scatter	1.26 %

Light curve

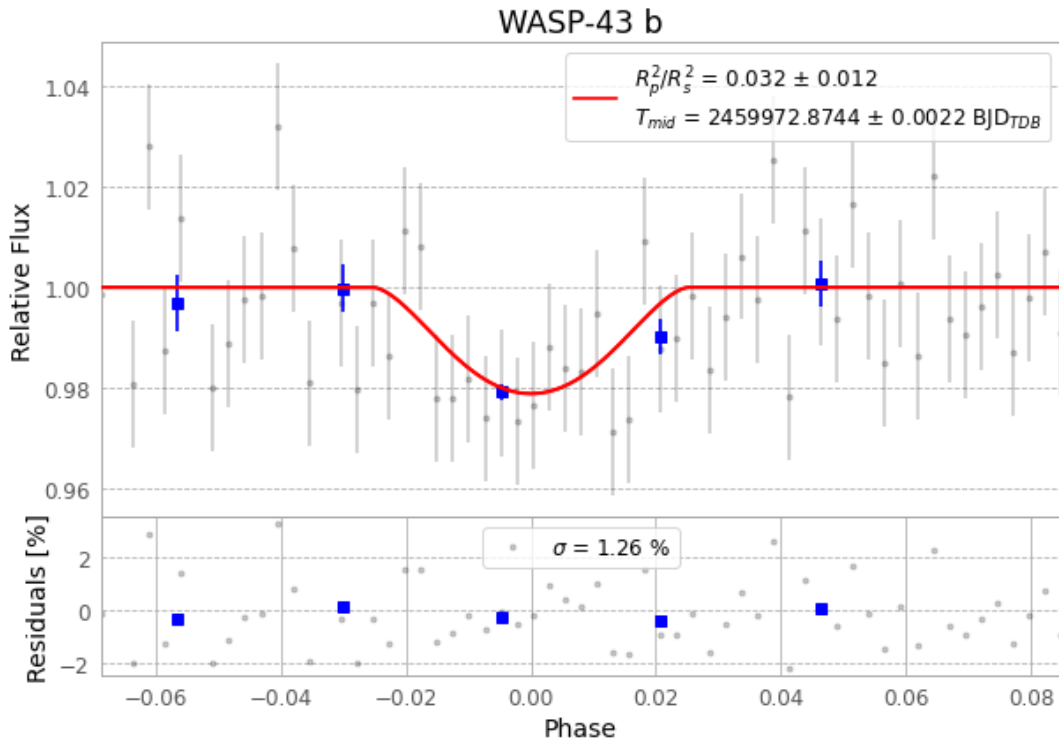


Figure 1 — FinalLightCurve_WASP-43 b_2023-01-28.png (SHA-256 6525a8d51cc9d917...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [446, 135], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 2a27740645f5b9ed...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

60 FITS frames (39 MB), WASP-43230128073015.FITS → WASP-43230128103015.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-230128.log (1dc164da9c44...), FinalLightCurve_WASP-43 b_2023-01-28.png (6525a8d51cc9...), FinalLightCurve_WASP-43 b_2023-01-28.csv (681ada8309c6...)

Compute resources

Wall time 100.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-20 18:58Z.